CIE AS PHYSICS (9702) • QUESTION PAPER

7.3 Doppler Effect for Sound Waves

Syllabus Section: 7. Waves **Paper Type:** Multiple Choice (Paper 1) **Total:** 14 Questions (14 Marks)

Question 1 Easy [1 mark]

Which of the following statements are true?

A When a source emitting a sound wave moves away from an observer, the observed frequency increases

B When a source emitting a sound wave moves away from an observer, the observed wavelength increases

C When a source emitting a sound wave moves towards an observer, the observed frequency decreases

D When a source emitting a sound wave moves towards an observer, the observed wavelength increases

Question 2 Easy [1 mark]

A source emits a sound wave of speed v and frequency f_s. The source moves towards an observer at a speed of v_s.

What is the observed frequency of the source?

A $\frac{f_s v}{v + v_s}$

B $\frac{f_s v_s}{v - v_s}$

C $\frac{f_s v}{v - v_s}$

D $\frac{f_s v_s}{v + v_s}$

Question 3 Easy [1 mark]

A train is moving towards a stationary observer. The train sounds its horn.

What are the correct changes in frequency, wavelength and pitch of the horn as heard by the observer?

	Frequency	Wavelength	Pitch
A.	higher	shorter	higher
B.	higher	shorter	lower
C.	lower	shorter	lower
D.	lower	longer	higher

A

B

C

D

Question 4 Easy [1 mark]

Which of the following equations is correct for the frequency of a source that is moving toward an observer?

A $f_s = \frac{f_0 v - v_s}{v}$

B $f_s = \frac{f_0 v + v_s}{v}$

C $f_s = \frac{f_o v}{v - v_s}$

D $f_s = \frac{f_o v}{v + v_s}$

Question 5 Easy [1 mark]

A car driving at a velocity of 10 m s^{-1} toward an observer sounds its horn with a frequency of 10 kHz. The speed of sound in air is 330 m s^{-1}.

What frequency does the observer hear?

A 8.9 kHz

B 9.7 kHz

C 10.1 kHz

D 10.3 kHz

Question 6 Medium [1 mark]

A police car siren emits a sound wave with a frequency f_s of 440 Hz. The car is travelling away from an observer at a speed of 30 m s^{-1}.

The speed of sound is 340 m s^{-1}.

Which of the following is the frequency the observer hears?

A 404 Hz

B 440 Hz

C 460 Hz

D 478 Hz

Question 7 Medium [1 mark]

Figure 1.1 shows object *P*, moving at velocity v_s. It emits a sound wave of frequency f_s. The speed of sound is *v*.

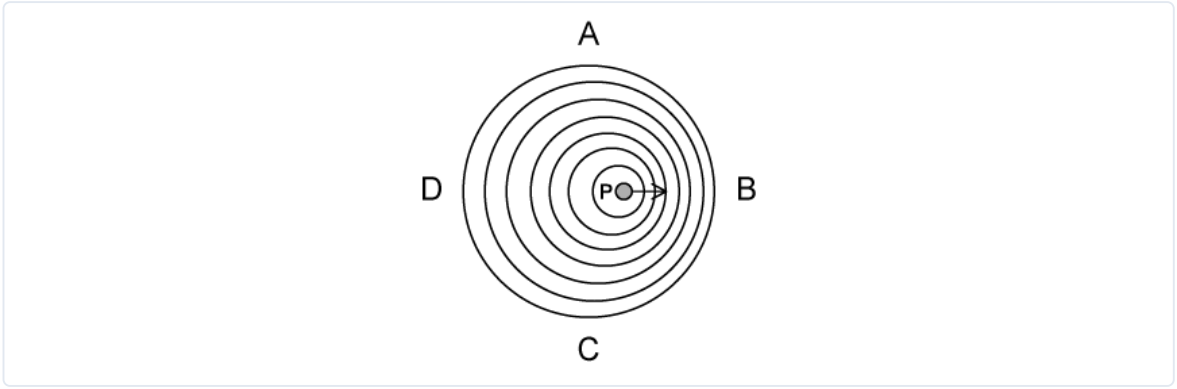


Fig. 1.1

At which point on the diagram will the observed frequency be $\frac{f_s v}{v + v_s}$?

- **A**
- **B**
- **C**
- **D**

Question 8 Medium [1 mark]

A source moving at constant velocity v_s, emits a sound wave of frequency 450 Hz. A stationary observer records a frequency of 300 Hz.

What is the value of v_s, and is it moving towards or away from the observer?

Speed of sound, v = 340 m s^{-1}

	v_s / m s^{-1}	Towards / Away
A.	170	Away
B.	170	Towards
C.	850	Away
D.	850	Towards

A

B

C

D

Question 9 Medium [1 mark]

A car of terrified observers try to get the car to start to drive away from a hungry T. rex, running toward them at a speed of 10 m s^{-1}. They hear the roar of the T. rex at a frequency of 170 Hz. The speed of sound in air is 340 m s^{-1}.

What is the frequency of the sound as emitted by the T. rex?

A 120 Hz

B 165 Hz

C 175 Hz

D 220 Hz

Question 10 Medium [1 mark]

A car travels in a straight line towards a stationary observer at a speed of 0.2v, where v is the speed of sound. The car's horn emits a pure sound of frequency f_s.

What is the frequency of the sound of the horn as heard by the observer, f_0?

A $f_0 = 0.83f_s$

B $f_0 = 1.00f_s$

C $f_0 = 1.20f_s$

D $f_0 = 1.25f_s$

Question 11 Hard [1 mark]

Two runners, A and B, run past each other in opposite directions, both at speed u. A police car has its siren on. It is traveling at a speed of $10u$, in the same direction as Runner A. Runner A and Runner B observe the sound wave from the siren at frequencies of f_A and f_B respectively. The speed of sound is v.

What is the ratio of $\frac{f_A}{f_B}$?

A $\frac{v - 11u}{v + 9u}$

B $\frac{v}{v + 12u}$

C $\frac{v - 11u}{v - 9u}$

D $\frac{v}{v - 10u}$

Question 12 Hard [1 mark]

Fig. 1.1 is a graph of the observed wavelength against time of a sound wave emitted from a single source. The emitted sound wave has a frequency f_s. The sound travels at a constant velocity directly towards an observer and a frequency of f_T is observed. It then travels directly away from the observer and a frequency of f_A is observed.

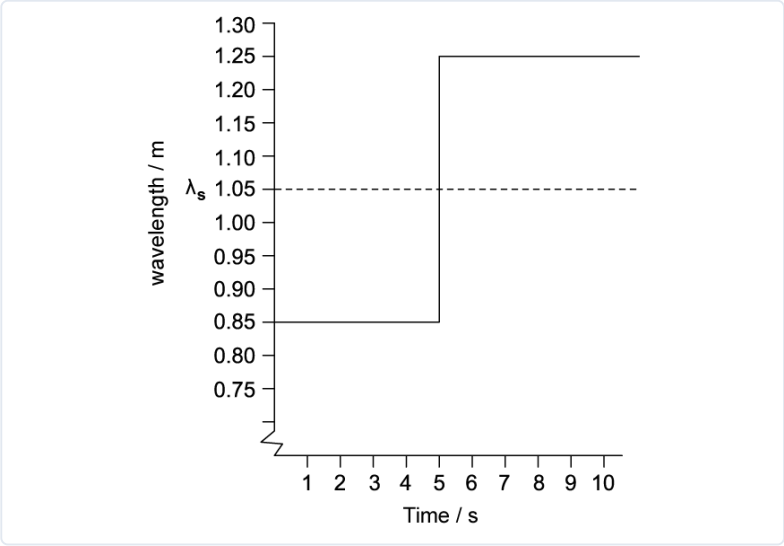


Fig. 1.1

What is the value of the ratio $\frac{f_A}{f_T}$?

A $\frac{25}{17}$

B $\frac{25}{21}$

C $\frac{17}{25}$

D $\frac{21}{25}$

Question 13 Hard [1 mark]

Fig 1.1 is a diagram of a source that has a mass of 500 g and is connected to a rope of length 20 cm. The rope is attached to point *Q* and has a tension of 800 N. The source travels around point Q anticlockwise in a circular path with a velocity of v_s. It emits a sound wave of frequency f_s. When the source is at point P, an observer stands directly in line to the direction of motion at point O.

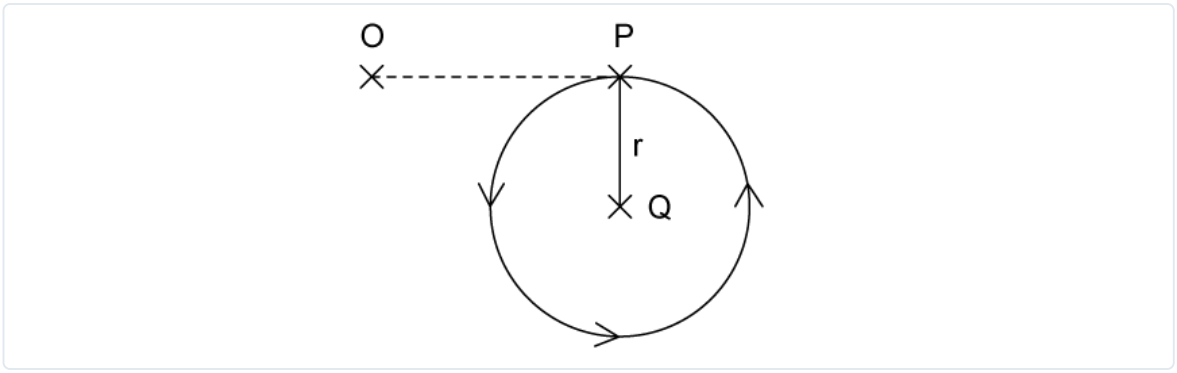


Fig 1.1

What is the value of v_s and $\frac{f_o}{f_s}$ when the source is at point P?

Speed of sound, v = 340 m s^{-1}

	v_s / m s^{-1}	$\frac{f_o}{f_s}$
A.	17.9	1.06
B.	17.9	0.950
C.	320	17.0
D.	320	0.515

A

B

C

D

Question 14 Hard [1 mark]

A source moves directly in line with an observer at a constant velocity v_s and emits a sound wave with a wavelength of λ_s. The observed wavelength of the source is λ_0 and the ratio $\frac{\lambda_0}{\lambda_s}$ is equal to $\frac{5}{4}$.

What is the value of v_s?

Speed of sound, v = 340 m s^{-1}

A 68.0 m s^{-1}

B 85.0 m s^{-1}

C 612 m s^{-1}

D 765 m s^{-1}